Mathematical Methods for Physics—Lecture 18

Physical Models, Data, and Initial-Value Problems

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We first derive the basic equations and explain their initial and boundary data. We then study initial-value problems on the whole line: travelling waves for the wave equation, followed by the Fourier-transform method for heat conduction. Finite-interval problems will be treated in the next lecture.

Throughout, $a > 0$ denotes wave speed and $\kappa > 0$ thermal diffusivity. The alternative heat-equation notation $a^2 = K/(\rho c_p)$ corresponds to κ here; it is distinct from the squared wave speed.

1 Three Basic Equations

For a scalar field $u = u(\mathbf{x}, t)$, the three-dimensional Laplacian is

$$\Delta u = u_{xx} + u_{yy} + u_{zz}. \tag{1}$$

The principal models are

$$u_{tt} - a^2 \Delta u = f(\mathbf{x}, t) \quad \text{(wave equation)}, \tag{2}$$

$$u_t - \kappa \Delta u = f(\mathbf{x}, t) \quad \text{(heat equation)}, \tag{3}$$

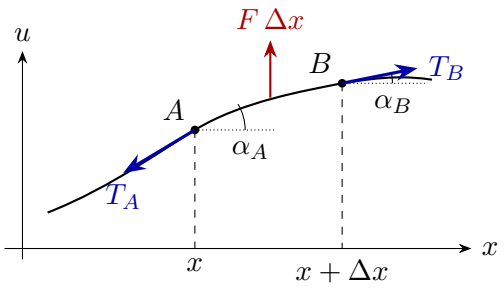
$$\Delta u = f(\mathbf{x}) \quad \text{(potential equation)}. \tag{4}$$

The source f has a different physical meaning in each model. When $f = 0$ the equation is homogeneous. The homogeneous potential equation is Laplace's equation, and its solutions are harmonic functions. With a source it is a form of Poisson's equation.

2 Derivation of the Equations

2.1 Small transverse vibrations of a string

Let $u(x, t)$ be the transverse displacement of a taut string. Its uniform reference mass per unit length is $\rho_s > 0$. Consider a small element with endpoints A and B, horizontal separation Δx, arc length Δs, endpoint tensions T_A, T_B, and tangent angles α_A, α_B.



Here $F(x, t)$ is an applied transverse force per unit horizontal length. For small transverse motion there is no horizontal acceleration to first order:

$$T_B \cos \alpha_B - T_A \cos \alpha_A = 0. \tag{5}$$

The small-slope approximation gives

$$|u_x| \ll 1, \qquad \cos\alpha \simeq 1, \qquad \sin\alpha \simeq \tan\alpha = u_x, \qquad \Delta s \simeq \Delta x. \tag{6}$$

Thus $T_A \simeq T_B = T$, a constant to this order. The transverse force balance is

$$T_B \sin\alpha_B - T_A \sin\alpha_A + F(x,t)\Delta x \simeq \rho_s \Delta x\, u_{tt}. \tag{7}$$

After division by Δx and passage to the limit,

$$\rho_s u_{tt} = T u_{xx} + F, \qquad \boxed{u_{tt} - a^2 u_{xx} = f, \quad a^2 = \frac{T}{\rho_s}, \quad f = \frac{F}{\rho_s}.} \tag{8}$$

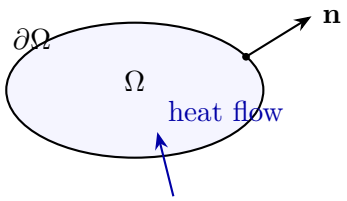
The cases $F = 0$ and $F \neq 0$ describe free and forced vibration. The analogous scalar membrane equation in two spatial dimensions uses $u_{xx} + u_{yy}$.

2.2 Heat conduction and energy conservation

Let u denote temperature, $K > 0$ thermal conductivity, $\rho > 0$ density, and $c_p > 0$ specific heat. Assume these coefficients are constant. Fourier's law is

$$\mathbf{q} = -K\nabla u. \tag{9}$$

On the surface of a fixed control volume Ω, let $\mathbf{n}$ be the *outward* unit normal. Then $\mathbf{q}\cdot\mathbf{n} = -K\partial_n u$ is the outward heat flux, so its negative is the inward heat flux.



If Q is heat generated per unit volume per unit time, conservation of energy over $[t_1, t_2]$ states

$$\int_\Omega \rho c_p \left[u(\mathbf{x}, t_2) - u(\mathbf{x}, t_1)\right] \mathrm{d}V = \int_{t_1}^{t_2} \left[-\int_{\partial\Omega} \mathbf{q}\cdot\mathbf{n}\, \mathrm{d}S + \int_\Omega Q\, \mathrm{d}V \right] \mathrm{d}t. \tag{10}$$

Use the fundamental theorem of calculus on the left and the divergence theorem on the right. Since the volume and time interval are arbitrary,

$$\rho c_p u_t = K\Delta u + Q, \qquad \boxed{u_t - \kappa\Delta u = f, \quad \kappa = \frac{K}{\rho c_p}, \quad f = \frac{Q}{\rho c_p}.} \tag{11}$$

The no-source derivation is recovered by setting $Q = 0$.

2.3 A stationary temperature field

If $u_t = 0$, steady heat conduction gives $K\Delta u + Q = 0$. In particular, a source-free stationary field satisfies $\Delta u = 0$. This connects a potential problem with the steady version of an evolution problem.

3 Initial and Boundary Data

3.1 Initial-value and initial-boundary-value problems

The equation alone does not select a physical solution. The heat equation needs one initial condition; the wave equation needs two:

$$u(\mathbf{x},0)=\varphi(\mathbf{x}) \qquad \text{(initial temperature or displacement)}, \tag{12}$$

$$u_t(\mathbf{x},0)=\psi(\mathbf{x}) \qquad \text{(initial velocity for the wave equation)}. \tag{13}$$

We first study *initial-value problems* on the whole line. On a finite interval, the initial data must be combined with endpoint conditions: this is an *initial-boundary-value problem*, also called a mixed problem here.

3.2 Three types of boundary conditions

With $\partial_n u=\nabla u\cdot\mathbf{n}$ and $\mathbf{n}$ outward, the three standard conditions are

Type	Prescribed quantity	Homogeneous case
Dirichlet	$u=g$	$u=0$
Neumann	$\partial_n u=g$	$\partial_n u=0$
Robin	$\partial_n u+\sigma u=g$	$\partial_n u+\sigma u=0$

For heat conduction these describe a prescribed surface temperature, a prescribed normal derivative (insulation when zero), or heat exchange with the surroundings. Newton cooling to temperature u_∞ gives

$$-K\partial_n u=h(u-u_\infty), \qquad \partial_n u+\frac{h}{K}u=\frac{h}{K}u_\infty, \quad h\ge 0. \tag{14}$$

For the transverse string model, $u=0$ describes a fixed end and $u_x=0$ a free transverse end. A restoring spring of stiffness k_s gives $T\partial_n u+k_s u=0$. On $0<x<\ell$, this means

$$-Tu_x(0,t)+k_s u(0,t)=0, \qquad Tu_x(\ell,t)+k_s u(\ell,t)=0. \tag{15}$$

The opposite signs arise because $\partial_n=-\partial_x$ on the left and $\partial_n=\partial_x$ on the right.

4 The Wave Equation on the Whole Line

4.1 Characteristic coordinates and travelling waves

Consider

$$u_{tt}-a^2u_{xx}=0, \qquad x\in\mathbb{R}. \tag{16}$$

Introduce $\xi=x+at$, $\eta=x-at$, and write $u(x,t)=U(\xi,\eta)$. The chain rule gives

$$u_x=U_\xi+U_\eta, \qquad u_t=a(U_\xi-U_\eta), \tag{17}$$

$$u_{xx}=U_{\xi\xi}+2U_{\xi\eta}+U_{\eta\eta}, \qquad u_{tt}=a^2(U_{\xi\xi}-2U_{\xi\eta}+U_{\eta\eta}). \tag{18}$$

Therefore $u_{tt}-a^2u_{xx}=-4a^2U_{\xi\eta}=0$. Integrating once in each characteristic coordinate yields

$$\boxed{u(x,t)=F(x+at)+G(x-at).} \tag{19}$$

Following a point of constant phase shows that $F(x+at)$ moves left and $G(x-at)$ moves right, both at speed a.

4.2 d'Alembert's formula

Impose $u(x,0)=\varphi(x)$ and $u_t(x,0)=\psi(x)$. Then

$$F(x)+G(x)=\varphi(x), \qquad F'(x)-G'(x)=\frac{\psi(x)}{a}. \tag{20}$$

If $H'=\psi$, the functions can be chosen as

$$F=\frac{\varphi}{2}+\frac{H}{2a}, \qquad G=\frac{\varphi}{2}-\frac{H}{2a}. \tag{21}$$

Substitution into (19) gives

$$\boxed{u(x,t)=\frac{\varphi(x+at)+\varphi(x-at)}{2}+\frac{1}{2a}\int_{x-at}^{x+at}\psi(s)\,\mathrm{d}s.} \tag{22}$$

For example, $\varphi\in C^2$ and $\psi\in C^1$ suffice for a classical solution. At (x,t) only the initial data in $[x-at,x+at]$ can affect the solution.

Example 11.1 (textbook p.298). For $\varphi(x)=x$ and $\psi(x)=\cos x$,

$$u(x,t)=x+\frac{\sin(x+at)-\sin(x-at)}{2a}=x+\frac{\cos x\sin(at)}{a}. \tag{23}$$

4.3 Nonhomogeneous waves and a particular solution

For $u_{tt}-a^2u_{xx}=f(x,t)$, the source-term extension is

$$\begin{aligned}u(x,t)=&\frac{\varphi(x+at)+\varphi(x-at)}{2}+\frac{1}{2a}\int_{x-at}^{x+at}\psi(s)\,\mathrm{d}s\\&+\frac{1}{2a}\int_0^t\int_{x-a(t-\tau)}^{x+a(t-\tau)}f(s,\tau)\,\mathrm{d}s\,\mathrm{d}\tau.\end{aligned} \tag{24}$$

The source response has zero initial displacement and velocity. Alternatively, find a particular solution W of the source equation and put $v=u-W$:

$$v_{tt}-a^2v_{xx}=0, \quad v(x,0)=\varphi(x)-W(x,0), \quad v_t(x,0)=\psi(x)-W_t(x,0). \tag{25}$$

Example 11.3 (textbook p.303). Solve

$$\begin{cases}u_{tt}-u_{xx}=-\sin t, & x\in\mathbb{R},\ t>0,\\ u(x,0)=\cos x, \qquad u_t(x,0)=e^x. & \end{cases} \tag{26}$$

The forcing depends only on time. Taking $W(t)=\sin t$ gives $W''=-\sin t$. Thus $v=u-\sin t$ has initial data $\cos x$ and e^x-1. With $a=1$,

$$v(x,t)=\frac{\cos(x+t)+\cos(x-t)}{2}+\frac{1}{2}\int_{x-t}^{x+t}(e^s-1)\,\mathrm{d}s \tag{27}$$

$$=\cos x\cos t+e^x\sinh t-t. \tag{28}$$

Consequently,

$$\boxed{u(x,t)=\cos x\cos t+e^x\sinh t-t+\sin t.} \tag{29}$$

Substitution verifies the PDE and both initial conditions.

5 Heat Conduction on the Whole Line

5.1 Fourier transform in space

We next consider heat conduction in an infinitely long rod:

$$\begin{cases} u_t - \kappa u_{xx} = f(x,t), \quad x \in \mathbb{R},\ t > 0, \\ u(x,0) = \varphi(x). \end{cases} \tag{30}$$

Recall the Fourier-transform convention from the preceding course unit:

$$\widehat{v}(\xi) = \int_{-\infty}^{\infty} v(x) e^{-i\xi x}\,\mathrm{d}x, \qquad v(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{v}(\xi) e^{i\xi x}\,\mathrm{d}\xi. \tag{31}$$

These inversion formulas are used for data with sufficient regularity and decay. Here ξ is a spatial frequency, and t is held as a parameter. Under conditions justifying integration by parts and differentiation under the integral,

$$\mathcal{F}_x[u_{xx}] = -\xi^2 \widehat{u}, \qquad \mathcal{F}_x[u_t] = \partial_t \widehat{u}. \tag{32}$$

Transform *both* the PDE and its initial condition:

$$\partial_t \widehat{u} + \kappa \xi^2 \widehat{u} = \widehat{f}(\xi,t), \qquad \widehat{u}(\xi,0) = \widehat{\varphi}(\xi). \tag{33}$$

For each ξ, this is a first-order ODE. The integrating factor $e^{\kappa\xi^2 t}$ gives

$$\widehat{u}(\xi,t) = e^{-\kappa\xi^2 t}\widehat{\varphi}(\xi) + \int_0^t e^{-\kappa\xi^2(t-\tau)}\widehat{f}(\xi,\tau)\,\mathrm{d}\tau. \tag{34}$$

5.2 Gaussian transform pair and the heat kernel

Recall the Gaussian transform pair

$$\mathcal{F}[e^{-\beta x^2}](\xi) = \sqrt{\frac{\pi}{\beta}} e^{-\xi^2/(4\beta)}, \qquad \beta > 0. \tag{35}$$

Thus

$$\mathcal{F}^{-1}[e^{-b\xi^2}](x) = \frac{1}{\sqrt{4\pi b}} e^{-x^2/(4b)}, \qquad b > 0. \tag{36}$$

Set $b = \kappa t$. The *heat kernel*, or fundamental solution, is

$$\boxed{G(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \exp\left(-\frac{x^2}{4\kappa t}\right), \quad t > 0.} \tag{37}$$

For positive time, $G_t - \kappa G_{xx} = 0$, $G > 0$, and $\int_{\mathbb{R}} G(x,t)\,\mathrm{d}x = 1$.

5.3 Inversion and the Poisson integral formula

Recall spatial convolution and its transform:

$$(v *_x w)(x) = \int_{\mathbb{R}} v(x-s) w(s)\,\mathrm{d}s, \qquad \mathcal{F}[v *_x w] = \widehat{v}\,\widehat{w}. \tag{38}$$

Invert (34) using (37):

$$u(x,t) = (G(\cdot,t) *_x \varphi)(x) + \int_0^t (G(\cdot, t-\tau) *_x f(\cdot,\tau))(x)\,\mathrm{d}\tau. \tag{39}$$

In full integral form,

$$u(x,t) = \int_{\mathbb{R}} \frac{e^{-(x-s)^2/(4\kappa t)}}{\sqrt{4\pi\kappa t}} \varphi(s)\,\mathrm{d}s + \int_0^t \int_{\mathbb{R}} \frac{e^{-(x-s)^2/[4\kappa(t-\tau)]}}{\sqrt{4\pi\kappa(t-\tau)}} f(s,\tau)\,\mathrm{d}s\,\mathrm{d}\tau. \tag{40}$$

The source integral uses the limit as $\tau \uparrow t$. For suitable data this limit is understood through the convolution. With $f = 0$, the formula becomes

$$\boxed{u(x,t) = \frac{1}{\sqrt{4\pi\kappa t}} \int_{\mathbb{R}} e^{-(x-s)^2/(4\kappa t)} \varphi(s)\,\mathrm{d}s.} \tag{41}$$

This is the Poisson integral formula for the heat equation (textbook p.309). As $t \downarrow 0$, it approaches the initial data at continuity points, or in an appropriate integral sense. For any $t > 0$ the Gaussian weights all spatial points; the wave equation instead has a finite interval of dependence.

From Initial-Value Problems to Mixed Problems

Both problems above are posed on $\mathbb{R}$, with no finite endpoints. On $0 < x < \ell$, the endpoints restrict the admissible spatial shapes. For example,

$$\sin(kx)\cos(akt) = \frac{1}{2}\{\sin[k(x+at)] + \sin[k(x-at)]\} \tag{42}$$

is a standing wave made from two counterpropagating waves. Fixed endpoints require $\sin(k\ell) = 0$, so only certain wave numbers are allowed. This leads to the eigenvalue problems and separation of variables in Lecture 19.